John **Gargalionis**

Postdoctoral Researcher · Particle Phenomenology
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Employment

University of Valencia
Valencia, Spain

Postdoctoral researcher — *Jan 2022*

- Studies of lepton- and baryon-number violation
- Delayed due to pandemic

University of Melbourne
Melbourne, Australia

Research Assistant — *May — Oct 2021*

- Developing novel techniques for quark–gluon discrimination using unsupervised machine learning and the CMS Open Data
- Extending doctoral work on connection between flavour anomalies and neutrino masses

Education

University of Melbourne
Melbourne, Australia

Doctor of Philosophy (Theoretical Particle Physics) — *2016 — 2020*

- *Thesis title:* Models of radiative neutrino mass and lepton flavour non-universality
- *Supervisor:* Prof. Raymond Volkas

Master of Science (with Distinction) — *2014 — 2016*

- *Thesis title:* Neutrino mass through leptoquarks: a new radiative model and its experimental prospects
- *Supervisors:* Prof. Raymond Volkas and Prof. Elisabetta Barberio

Bachelor of Science — *2010 — 2014*

- *Specialisations:* Physics, Neuroscience, Ancient Languages

Publications

Research publications

Exploding operators for Majorana neutrino masses — *2021*
John Gargalionis & Raymond R. Volkas — *JHEP*
`arXiv:2009.13537, JHEP 01 (2021)`

Radiative neutrino mass model from a mass dimension 11 $\Delta L = 2$ effective operator — *2020*
John Gargalionis, Iulia Popa-Mateiu & Raymond R. Volkas — *JHEP*
`arXiv:1912.12386, JHEP 03 (2020)`

A near-minimal leptoquark model for reconciling flavour anomalies and generating radiative neutrino mass — *2019*
Innes Bigaran, John Gargalionis & Raymond R. Volkas — *JHEP*
`arXiv:1906.01870, JHEP 10 (2019)`

Reconsidering the One Leptoquark solution: flavour anomalies and neutrino mass — *2017*
Yi Cai, John Gargalionis, Michael A. Schmidt & Raymond R. Volkas — *JHEP*
`arXiv:1704.05849, JHEP 10 (2017)`

Explaining the 750 GeV diphoton excess with a coloured scalar charged under a new confining gauge interaction — *2016*
Robert Foot & John Gargalionis — *PRD*
`arXiv:1604.06180, Phys. Rev. D94:011703 (2016)`

Other publications

Solutions to Problems at Les Houches Summer School on EFT — *2020*

Marcel Balsiger, Marios Bounakis, Medhi Drissi, John Gargalionis, Erik Gustafson, Greg Jackson, Matthew Leak, Christopher Lepenik, Scott Melville, Daniel Moreno, Michele Tommaro, Selim Touati & Timothy Trott

Oxford University Press (included in publication of lecture notes)

`arXiv:2005.08573`

General Skills

Programming
Python (SciPy, NumPy, SymPy, Jupyter, etc.), Mathematica (Wolfram Language), C/C++, Clojure(Script), Scheme, Haskell
- Practised in test-driven software development and version control using git
- Experience working with high-performance computing clusters using PBS and slurm

Data analysis
ROOT, SciPy, Pandas, Dask, h5py, SQLite

Machine learning
TensorFlow, Keras, (Un)supervised learning, Image classification

Communication
Public speaking, scientific writing, science communication

Academic Skills

Model building and model exploration
Radiative neutrino mass, group theory, flavour phenomenology, GUTs
- Constructing and analysing extensions of the Standard Model, experience mostly in models of radiative neutrino mass
- Expertise in the group theory of $\mathrm{SU}(N)$ and $\mathrm{SO}(N)$ in the context of GUT model building
- Knowledgable about new-physics explanations of the B-meson anomalies and their connection to other topics in flavour
- Confident in analysing the viability of BSM models, especially in the context of flavour and collider constraints

Effective Field Theory
SMEFT, WET, $\Delta L = 2$ EFT
- Worked extensively with the Standard Model EFT at mass-dimension 6 and higher
- On top of current developments in operator-basis generation (i.e. Hilbert series and computational techniques)

Collider phenomenology
ROOT, CMSSW, MadGraph, Pythia, Fastjet, Delphes
- Confident in the generation of Monte-Carlo data starting from a Lagrangian
- Experience with the CMS analysis framework through the use of their publicly available data
- Well acquainted with concepts in experimental analysis, experience with a range of physics objects and signatures

Calculation and fitting
LieArt, Package-X, FeynRules, FeynArts, FormCalc, LoopTools, Flavio
- Calculating loop-level amplitudes by hand and with Mathematica packages (both analytically and numerically)
- Confident with the use of Flavio for flavour calculations and fits

Invited Seminars and Short Stays

May 2022	**Laboratory of Subatomic Physics & Cosmology (LPSC)**, Two-week research visit	*Grenoble, France*
Oct 2018	**Belle II Theory Interface Platform**, Leptoquarks and flavour	*KEK, Japan*
May 2018	**Monash University**, Radiative neutrino mass and the flavour anomalies: a circumstantial case	*Melbourne, Australia*
May 2017	**Instant workshop on B-meson anomalies**, Reconsidering the 'one leptoquark' solution: flavour anomalies and neutrino mass, https://cds.cern.ch/record/2265323	*CERN, Switzerland*

Contributed Seminars

Sep 2022	**Korean Institute for Advanced Study (KIAS)**, Exploding Operators for Majorana neutrino masses	*Seoul, Korea*
June 2022	**HEFT 2022**, Exploding Operators for Majorana neutrino masses	*Granada, Spain*
May 2022	**Instituto de Física Corpuscular (IFIC)**, Exploding Operators for Majorana neutrino masses	*Valencia, Spain*
Nov 2020	**PhD Completion Seminar**, Models of radiative neutrino mass and lepton-flavour non-universality	*Melbourne, Australia*
Sep 2017	**Geoff Opat Seminar Series**, Radiative neutrino mass and the flavour anomalies: a circumstantial case	*Melbourne, Australia*
Aug 2017	**Technische Universität Dortmund**, Radiative neutrino mass and the flavour anomalies: a circumstantial case	*Dortmund, Germany*
Aug 2017	**Technische Universität München**, Radiative neutrino mass and the flavour anomalies: a circumstantial case	*Garching, Germany*
Dec 2016	**APPC-AIP Congress**, Reconsidering the 'one leptoquark' solution: flavour anomalies and neutrino mass	*Brisbane, Australia*
Jun 2016	**University of Melbourne**, Light leptoquarks at the LHC: neutrino mass and flavour physics	*Melbourne, Australia*
Nov 2015	**MSc Completion Seminar**, Radiative neutrino mass through leptoquarks	*Melbourne, Australia*

Training

Summer schools

Aug 2017	**Joint Challenges for Cosmology and Colliders**, MITP	*Mainz, Germany*
Jul 2017	**EFT in Particle Physics and Cosmology**, Ecole de Physique des Houches	*Les Houches, France*
Jul 2016	**Pre-SUSY School**, University of Melbourne	*Melbourne, Australia*

Summer research projects

Dark matter and heavy-flavoured quarks *Melbourne, Australia*

ATLAS EXOTICS GROUP *Jan 2016*

- *Supervisor:* Dr. Francesca UNGARO

Awards

Fellowships

Juan de la Cierva Fellowship *Valencia, Spain*

INSTITUTO DE FÍSICA CORPUSCULAR (IFIC) *2022*

- Total score: 90.10%

Scholarships & Honours

2021	**Chancellor's prize**, University of Melbourne	*Melbourne, Australia*
2018	**Science Abroad Travel Scholarship**, University of Melbourne	*Melbourne, Australia*
2017	**Research poster award**, CoEPP	*Adelaide, Australia*
2016 – 2019	**Australian Postgraduate Award (APA)**, Australian Research Council	*Melbourne, Australia*
2015	**Prof. Kernot Research Scholarship in Physics**, University of Melbourne	*Melbourne, Austria*
2014	**N. D. Goldsworthy Scholarship**, University of Melbourne	*Melbourne, Austria*

Teaching and Supervision

2022	**Masters Supervision**	*University of Valencia*
2020, 2021	**Teaching assistant**, Quantum Field Theory	*University of Melbourne*
2020	**High School Teacher**, VCE Modern Greek	*Victorian School of Languages*
2018	**Tutor**, 1st year Physics	*University of Melbourne*
2018, 2019	**Tutor**, 3rd year Quantum Mechanics	*University of Melbourne*
2018	**Tutor**, Advanced Scientific Programming in Python (Asia–Pacific)	*Melbourne Bioinformatics*
2016 – 2019	**Teaching assistant and course content creator**, 3rd year Subatomic Physics	*University of Melbourne*
2014 – 2016	**Curriculum designer and language teacher**, Modern and Ancient Greek	*Centre for Adult Education*
2014 – 2018	**Laboratory demonstrator**, 1st year Physics Lab, 3rd year Particle Lab, 2nd year Computational Physics	*University of Melbourne*
2010 – 2019	**Private Tutor**, Physics, Mathematics, Greek	

Scientific Outreach

2018, 2019	**CoEPP Work Experience Program**, Introductory talk: The Standard Model	*University of Melbourne*
2017	**Physics Workshops**, Project coordinator	*Hume Central Secondary*
2016	**Undergraduate Seminar Series**, Research talk targeted at undergraduate physics students	*University of Melbourne*
2015, 2016	**International Masterclass in Particle Physics**	*South Oakleigh Grammar*
2015	**CoEPP Work Experience Program**, Organising committee	*University of Melbourne*
2015	**International Masterclass in Particle Physics**	*University of Melbourne*

Other

Citizenship Australian
Birth year 1991
Languages English (native), Greek (C1), Spanish (B1)